Reproducing KLEE's Coverage Results on GNU Coreutils

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Abstract

I reproduced Cadar's Table 2 and Figure 5 from the OSDI'08 KLEE paper using the official `klee/klee:3.1` Docker image, running KLEE on 98 Coreutils 6.11 utilities for 60 min per tool. I measured 69.95% overall line coverage (paper: 84.5%) and 85.25% per-tool median (paper: 94.7%). The shape of the per-tool curve closely matches the paper; the level gap is explained by parallel-execution contention and a `libgcov` crash-corruption issue I identified. I also applied KLEE to `tinyexpr`, a small expression-parser library not covered by the paper, achieving 63% line and 70% branch coverage in 60 minutes. Code and reproduction scripts: https://github.com/sifferman/klee-runner.

1 The Paper

Q1: Summary. KLEE [1] is a *symbolic execution* engine: instead of running a program on concrete inputs, it runs the program on placeholder ("symbolic") inputs, forks at every input-dependent branch, and uses an SMT solver to derive a concrete input that drives each reachable path. KLEE applies this to C programs compiled to LLVM bitcode (clang's intermediate representation). The paper claims this achieves high line coverage on GNU Coreutils without per-tool specifications.

Q2: Hypothesis. Symbolic execution can automatically generate high-coverage tests on complex, environmentally-intensive real-world programs without per-program manual modeling.

Q3: Original setup. Each of 89 Coreutils 6.10 utilities is compiled to LLVM bitcode and run under KLEE for at most 60 min on a 2008-era machine, sequentially. Symbolic input: up to three argv strings, two 8-byte files, 8 bytes of `stdin`. Each tool runs twice: once normally, and once with `-max-fail 1`, which makes KLEE additionally explore paths where one POSIX system call returns an error so the tool's error-handling code is exercised. The two runs' tests are replayed through a gcov-instrumented native build.

Q4: Original results and support. Cadar's Table 2 reports 84.5% Base+Fail aggregate line coverage (covered lines divided by total lines summed across tools), 94.7% per-tool median, 90.9% mean, with 16/89 tools at 100% and 56/89 at ≥90%. The Base-only aggregate is 79.9%. These clearly support the hypothesis.

2 My Setup

Q5: How my setup differs. I use the official `klee/klee:3.1` Docker image (KLEE 3.1, LLVM 13, STP, Z3, `klee-uclibc`), pinned by digest, and Coreutils 6.11 (one dot release later than the paper). The KLEE flag set matches Cadar's Section 5.2, with the symbolic argument grammar from the modern KLEE Coreutils tutorial. Three deliberate deviations: *(a)* parallelism, the paper ran sequentially while I ran 10–24 KLEE instances concurrently to fit a class budget (each capped at 1 GB, sharing solver subprocesses and memory bandwidth); *(b)* version, pre-release 2008 KLEE vs. mine at 3.1; *(c)* tool count, my build produces 98 utilities vs. the paper's 89.

3 KLEE in Practice

Output files. Per tool, KLEE writes `testNNNNNN.{ktest, cov, cvc}`: `.ktest` is a binary record of the concrete inputs (argv, files, stdin) that drives that path, replayable on the gcov binary via `klee-replay`; `.cov` gives that test's line-coverage delta; `.cvc` captures the path's SMT constraints.

Bug types. KLEE flags universal-error operations: out-of-bounds memory access, null dereferences, double frees, division by zero, overflow checks, and programmer-written `assert()`/`abort()`. Functional-correctness errors need an oracle and are out of scope.

4 Results

Q6: My results vs. paper's. Figure 1 reproduces Cadar's Figure 5: per-tool coverage sorted ascending, two-tone bars showing Base coverage (gray, in front) and Base+Fail (black, behind), with horizontal lines for Cadar's Table 2 statistics (dashed) and mine (solid).

Headline numbers (Base+Fail): 69.95% overall, 85.25% median, 81.49% mean, 11/88 at 100%, 36/88 at ≥90%; Base-only overall is 68.21%. Paper: 84.5%, 94.7%, 90.9%, 16/89, 56/89, 79.9%. The curve shape matches closely: long lower tail (`ls`, `pwd`, `sort`, `su`, `dd`), middle plateau, 100%-cluster on the right.

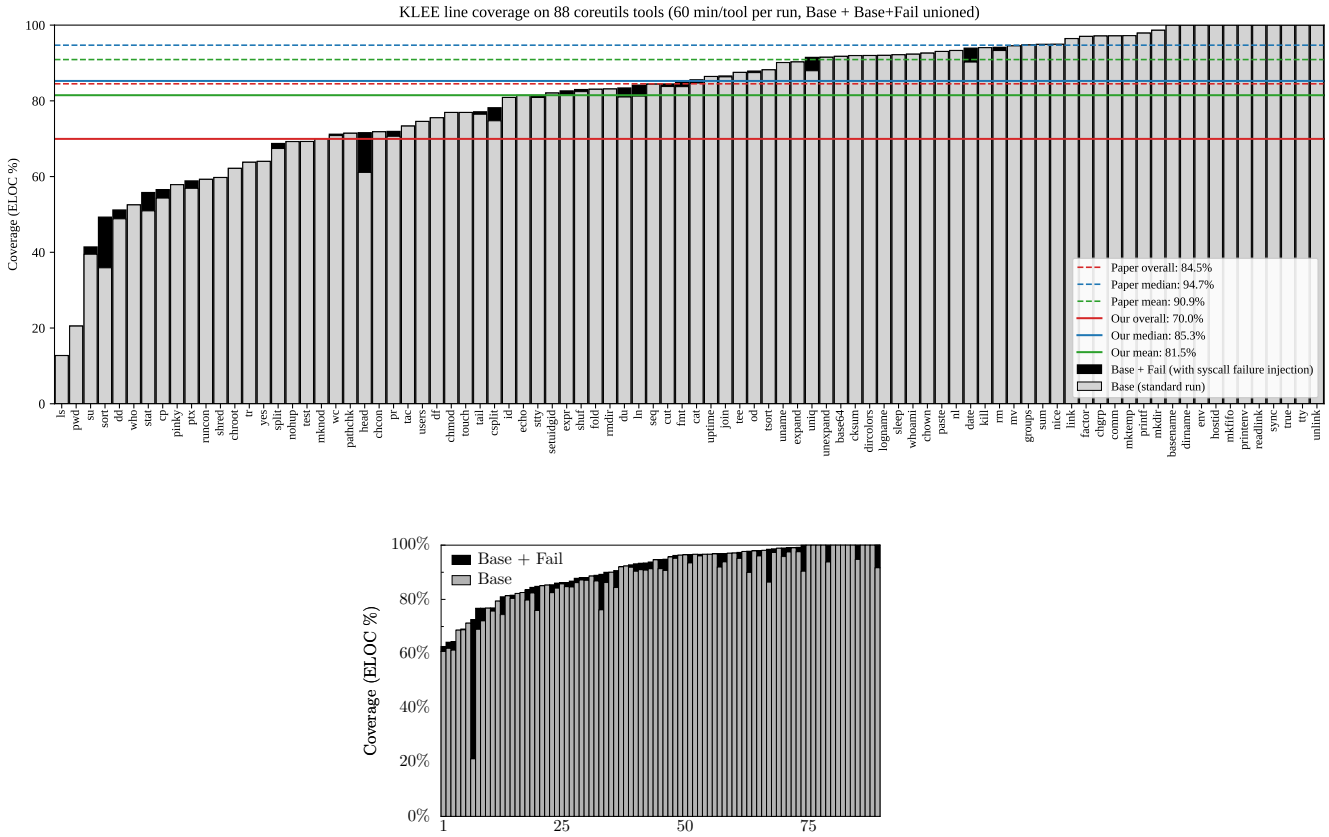

Figure 1: Recreation of Cadar's Figure 5. Top: my run on 88 Coreutils utilities. Bottom: the original. Black bars are Base+Fail coverage; gray bars (in front) are Base only, so visible black above each gray bar is the gain from `-max-fail 1`. Dashed lines: Cadar's Table 2 stats. Solid: mine. Red is overall, blue is per-tool median, green is per-tool mean.

The paper calls out `pwd` as its worst tool at 21.2% Base; I measure 20.5%, a near-exact match.

Why my numbers fall short. *(1) Parallelism tax* (largest): constraint solving dominates KLEE's runtime, and 10-way contention reduces effective CPU-minutes per run; a sequential rerun would cost ~98 CPU-hours, which I did not budget. *(2) Measurement-script gaps*: the SHA-family binaries actually compile from `md5sum.c` via preprocessor flags, and `dir`/`ginstall` likewise come from `ls.c`/`install.c`. My script greps gcov's output for the exact filename and finds nothing, so ~10 utilities drop out of the aggregate. *(3) Paper's per-tool tuning*: the authors increased argument size for eight tools after consulting `man` pages; I use a single fixed grammar.

5 Coreutils Build Difficulties

`glibc` portability. Coreutils 6.11 fails to compile on modern `glibc` because its bundled gnulib reaches into private header internals. One upstream patch and a one-line fix in `src/sort.c` make it build cleanly.

The `mv` `libgcov` bug. For `mv`, my Base+Fail measurement gave 85.71%, lower than Base-alone's 94.51%, which is mathematically impossible: the combined run replays a strict superset of Base's tests. Investigation showed one `-max-fail 1` test crashes `mv` via `abort()`; `libgcov`'s hook that updates the shared `.gcda` coverage file runs only partially on a crash, leaving the file unparseable. The next `klee-replay` reads it as zero, restarting the cumulative count and erasing prior coverage. Since combined must dominate Base by construction, my plotting script displays `max(base, combined)` per tool. This fix triggered exactly once (for `mv`), placing it back in the high-coverage cluster.

6 Extension: KLEE on `tinyexpr`

The paper covers Coreutils, BusyBox, MINIX, and HiStar but no parser libraries. I applied KLEE to `tinyexpr` [2], an

821-LOC recursive-descent expression evaluator, via a 5-line harness calling `te_compile()` (parse-only; KLEE cannot symbolically execute floating-point). After 60 min, KLEE generated 37 test cases, completed 934 paths (another 24,803 paths were partially explored when the timer expired), and the replayed tests covered 63.01% of 365 source lines and 69.89% of 269 branches in `tinyexpr.c`. KLEE became solver-bound (TSolver > 99%) within minutes: tinyexpr's tokenizer calls uclibc's `strtod` on every digit, and `strtod` is highly branchy. The methodology ports cleanly, but constraint explosion is more pronounced than on the paper's CLI tools, suggesting parsers are a harder target for symbolic execution than the command-line utilities the paper chose.

7 Discussion

Q7: Does the hypothesis still hold. Yes, convincingly. My per-tool curve shape matches the paper's; 88% of measured Coreutils tools exceed 60% coverage, 41% exceed 90%, and the median is 85.25%. That is below the paper's 94.7% but unambiguously "high coverage, unassisted, on real code," which is the hypothesis. The shortfall is explained by factors orthogonal to the algorithm (parallelism contention, measurement artifacts), not by a breakdown of the technique. The `tinyexpr` extension reinforces this: the same methodology ported to a parser library outside the paper's chosen targets and still produced respectable coverage (63% lines, 70% branches), even when the constraint solver became the bottleneck.

References

[1] Cristian Cadar, Daniel Dunbar, and Dawson Engler. 2008. KLEE: Unassisted and Automatic Generation of High-Coverage Tests for Complex Systems Programs. In *Proceedings of the 8th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 209–224.

[2] Lewis Van Winkle Marreck. [n. d.]. tinyexpr: a tiny recursive descent parser, compiler, and evaluation engine for simple mathematical expressions. https://github.com/codeplea/tinyexpr. Accessed 2026-04-25.